Lecture-10

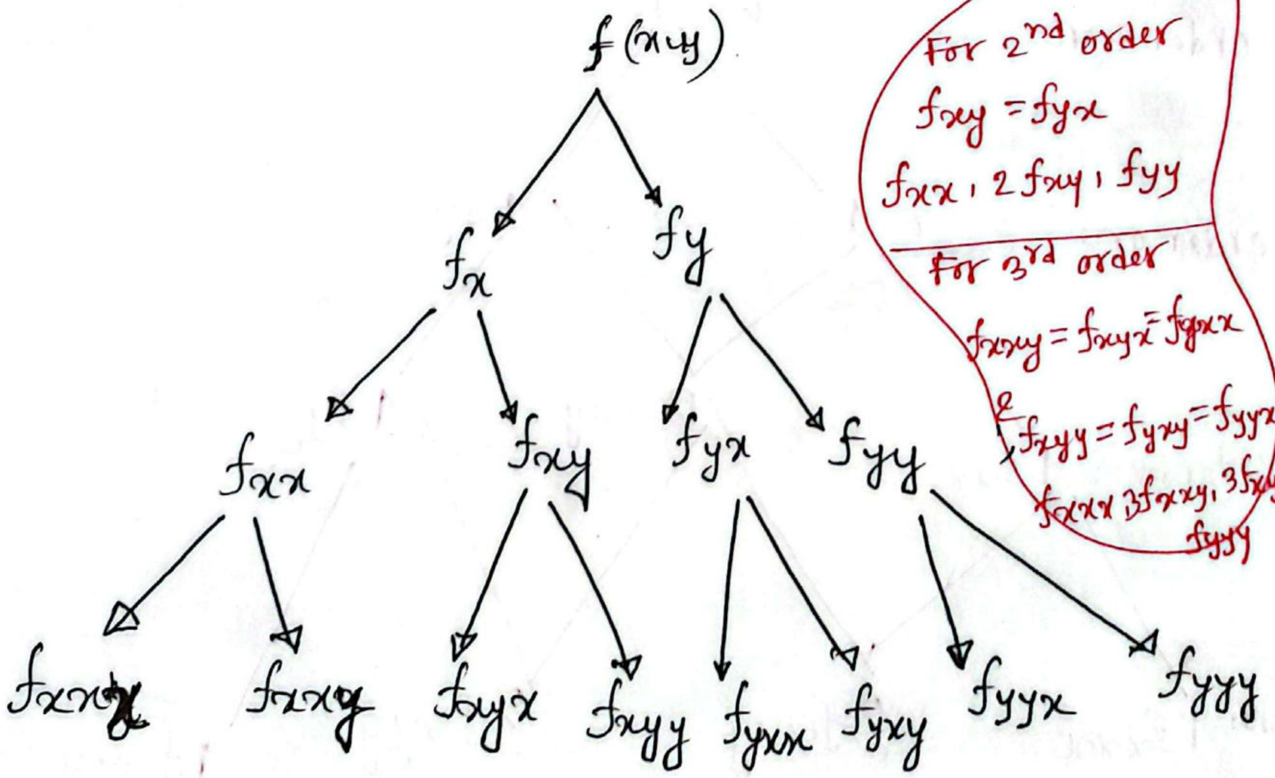
* Clairaut's theorem:

Suppose that f is defined on a Disk D that contains the point (a,b). If the function f_{xy} and f_{yx} are continuous on the Disk then,

$$f_{xy}(a,b) = f_{yx}(a,b).$$

Similarly,

$$f_{xyx} = f_{yxx}$$



For 2^{nd} order

$f_{xy} = f_{yx}$

$f_{xx},\ 2f_{xy},\ f_{yy}$

For 3^{rd} order

$f_{xxy} = f_{xyx} = f_{yxx}$

$f_{xyy} = f_{yxy} = f_{yyx}$

$f_{xxx},\ 3f_{xxy},\ 3f_{xyy},\ f_{yyy}$

Let $f(s, r, t)$ be a function of s, r, and t. then

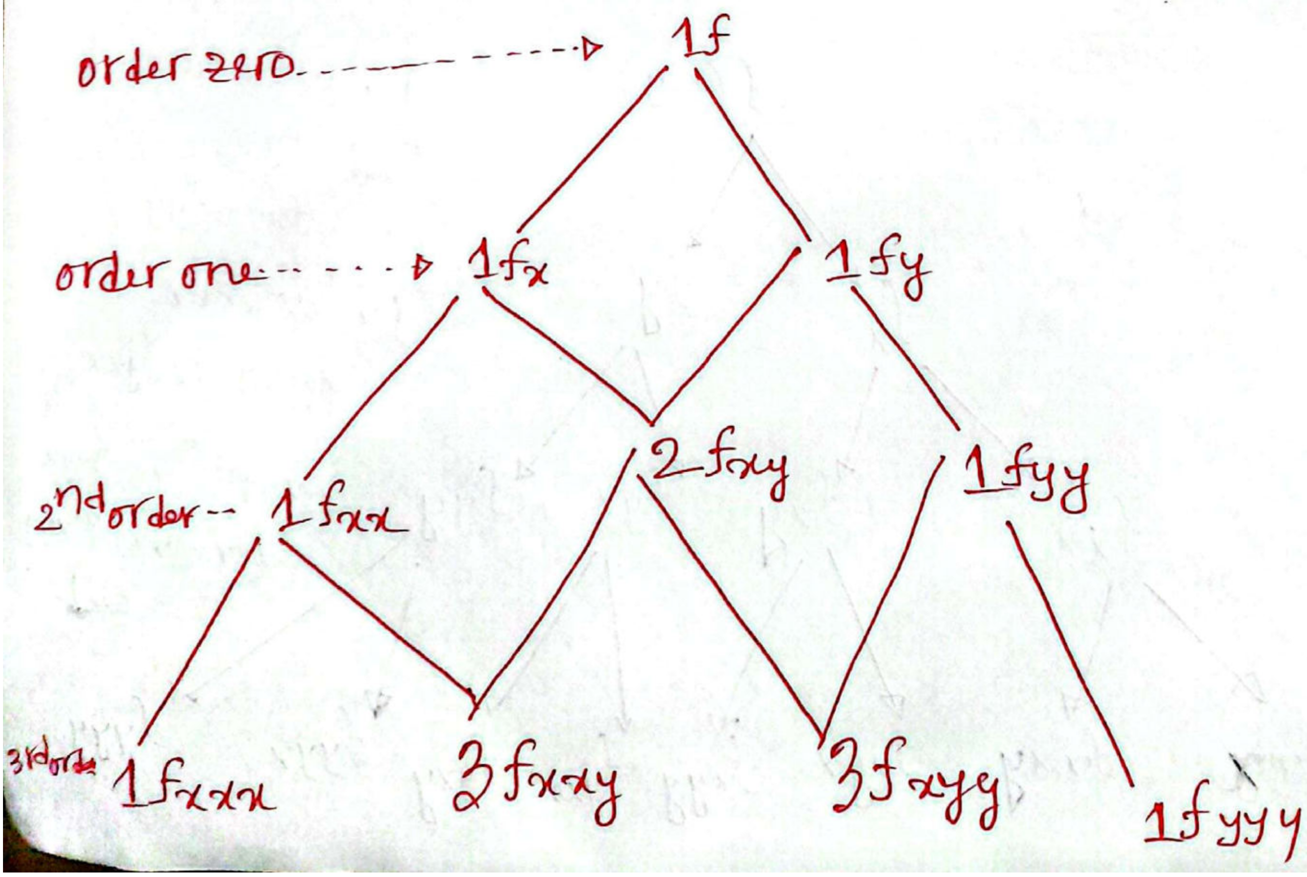
According to Clairaut's theorem

$$f_{ssrtsrr} = f_{trsrssr}$$

1 time by t
3 time by s
3 time by r

Because in each case you differentiate with respect to t once, s three times and r three times.

★ Tree for Taylor series expansion of $f(x,y)$



* Taylor series Expansion for two variables:

* The taylor series of a real or complex function $f(x)$ that is infinitely differentiable in a neighborhood of a real or complex number a is the power series

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \frac{f^{iv}(a)}{4!}(x-a)^4 + \cdots \cdots$$

* Taylor series for ~~done~~ two variables $f(x,y)$ at (a,b) is

$$f(x,y) = f(a,b) + \frac{f_x(a,b)}{1!\,0!}(x-a)^1(y-b)^0 + \frac{f_y(a,b)}{0!\,1!}(x-a)^0(y-b)^1 + \frac{f_{xx}(a,b)}{2!}(x-a)^2 + \frac{2f_{xy}(a,b)}{2!}(x-a)(y-b) + \frac{f_{yy}(a,b)}{2!}(y-b)^2 + \frac{f_{xxx}(a,b)}{3!}(x-a)^3 + \frac{3f_{xyy}(a,b)}{3!}(x-a)(y-b)^2 + \frac{3f_{xxy}(a,b)}{3!}(x-a)^2(y-b) + \frac{f_{yyy}(a,b)}{3!}(y-b)^3 + \cdots\cdots$$

$$\therefore f_0(x,y) = f(a,b) + \frac{1}{1!}\left[f_x(a,b)(x-a) + f_y(a,b)(y-b)\right]$$

$$+ \frac{1}{2!}\left[f_{xx}(a,b)(x-a)^2 + 2f_{xy}(a,b)(x-a)(y-b) + f_{yy}(a,b)(y-b)^2\right]$$

$$+ \frac{1}{3!}\left[f_{xxx}(a,b)(x-a)^3 + 3f_{xxy}(a,b)(x-a)^2(y-b)\right.$$

$$\left. + 3f_{xyy}(a,b)(x-a)(y-b)^2 + f_{yyy}(a,b)(y-b)^3\right]$$

$$+ \cdots\cdots\cdots .$$

When you ~~ask~~ are asked to calculate the polynomial, you need not put the dot sign. Because the dot sign is used to notify the series expansion.

* 1^{st} degree taylor polynomial of $f(x,y)$ near the point (a,b)

$$P_1(x,y) = L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

Here $L(x,y)$ is called the linear (or tangent plane) approximation of f for (x,y) near the point (a,b)

* 2^{nd} degree taylor polynomial

$$P_2(x,y) = Q(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$
$$+ \frac{1}{2!}\left[f_{xx}(a,b)(x-a)^2 + 2 f_{xy}(a,b)(x-a)(y-b) + f_{yy}(a,b)(y-b)^2\right]$$

* 3^{rd} Degree Taylor polynomial

$$P_3(x,y) = C(x,y) = Q(x,y) + \frac{1}{3!}\Big[f_{xxx}(a,b)(x-a)^3 + 3 f_{xxy}(a,b)(x-a)^2(y-b)$$
$$+ 3 f_{xyy}(a,b)(x-a)(y-b)^2 + f_{yyy}(a,b)(y-b)^3\Big]$$

$z_3 + P_2 h$

* Example: calculate the second degree taylor polynomial of the function $f(x,y) = \sin 2x + \cos y$ near $(0,0)$.

solution

Given that

$$f(x,y) = \sin 2x + \cos y \quad ; f(0,0) = 1$$

1st order partial derivatives

$$f_x(x,y) = 2\cos 2x \quad ; f_x(0,0) = 2$$

$$f_y(x,y) = -\sin y \quad ; f_y(0,0) = 0$$

2nd order partial derivatives

$$f_{xx} = -4 \sin 2x \quad ; \quad f_{xx}(0,0) = 0$$

$$f_{xy} = 0 = f_{yx} \quad ; \quad f_{xy}(0,0) = 0$$

$$f_{yy} = -\cos y \quad ; \quad f_{yy}(0,0) = -1$$

Therefore, the 2nd ~~order~~ ~~partial~~ degree Taylor polynomial of $f(x)$ near $(0,0)$ is

$$g(x,y) = 1 + \frac{1}{1!}\left(2(x-0) + 0(y-0)\right) + \frac{1}{2!}\left(0(x-0)^2 + 0(x-0)(y-0) + (-1)(y-0)^2\right)$$

$$= 1 + 2x - \frac{y^2}{2}$$

$$f(x,y) \approx 1 + 2x - y^2/2$$

* Expand $x^2y + 3y - 2$ in powers of $(x-1)$ and $(y+2)$.

Ans: let use ~~expand~~ Taylor's expansion with $a=1$; $b=-2$; near $(1,-2)$

$f(x,y) = x^2y + 3y - 2$; $f(1,-2) = -10$

1st order,

$f_x(x,y) = 2xy$; $f_x(1,-2) = -4$

$f_y(x,y) = x^2 + 3$; $f_y(1,-2) = 4$

P2k

2nd order...

$f_{xx}(x,y) = 2y \quad ; \quad f_{xx}(1,-2) = -4$

$f_{xy}(x,y) = 2x \quad ; \quad f_{xy}(1,-2) = 2$

$f_{yy}(x,y) = 0 \quad ; \quad f_{yy}(1,-2) = 0$

3rd order

$f_{xxx} = 0 \quad ; \quad f_{xxx}(1,-2) = 0$

$f_{xxy} = 2 \quad ; \quad f_{xxy}(1,-2) = 2$

$f_{xyy} = 0 \quad ; \quad f_{xyy}(1,-2) = 0$

$f_{yyy} = 0 \quad ; \quad f_{yyy}(1,-2) = 0$

All higher derivatives are zero.

$$\therefore f(x,y) = -10 + \frac{1}{1!}\left(-4(x-1) + 4(y+2)\right)$$
$$+ \frac{1}{2!}\left(-4(x-1)^2 + 2\times 2(x-1)(y+1) + 0(y-2)^2\right)$$
$$+ \frac{1}{3!}\left(0 + 3\times 2(x-1)^2(y-1)^1 + 3\times 0 + 0\right)$$

$$f(x,y) = -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + (x-1)^2(y+2)$$ Ans.

problem: Expand the function $f(x,y) = x^3 + 3x^2y^2 + 4xy^2 + y^3$ by Taylor's theorem in powers of $(x-1)$ & $(y-1)$.